

Sheet 4 – sol

Q1 - If the neighborhood is too small, only the winner neuron updates or only a few neurons around it update. This can cause:

- poor global organization
- many neurons staying unused
- fragmented or noisy maps
- loss of topology preservation

Why: the neighborhood is what spreads learning across nearby neurons. Without enough neighborhood effect, the map does not organize smoothly and behaves more like isolated competitive learning rather than a coherent self-organizing map.

Q2 - Dimensionality reduction means converting data with many features into a lower-dimensional representation, often a 2D map, while preserving important similarity relationships.

For example, an input vector may have 50 dimensions, but after training a SOM, similar data points are placed close together on a 2D grid.

Practical application:

In customer segmentation, each customer may be described by age, income, spending habits, product preferences, and other features. A SOM can place similar customers in nearby regions of a 2D map, helping businesses visualize groups and target them more effectively.

Q3- SOMs are interesting because they mimic some principles observed in biological neural systems:

- neurons compete to respond
- nearby neurons cooperate
- spatially ordered maps emerge automatically
- similar stimuli are represented by nearby neurons

This is similar to how sensory areas in the brain organize responses to touch, sound, and vision.

Q4- Each of the k neurons has a weight vector of dimension m .

So the total number of weights is:

$$k \times m$$

Q5 - The competitive process works as follows:

1. Present an input vector to the network.
2. Compute the similarity or distance between the input and each neuron's weight vector.

3. Select the neuron whose weight vector is closest to the input.
4. This neuron is the winner or BMU.

Usually the winner is found by minimizing Euclidean distance.

Q6- It learns the **structure of the input data** rather than target labels. This includes:

- clusters
- similarities
- correlations
- distribution of patterns
- representative prototypes

So unsupervised learning discovers how the data is organized.

Q7- A basic SOM has **two layers**:

1. Input layer

- receives the input vector
- one input node per feature

2. Output/map layer

- consists of neurons arranged in a 1D or 2D grid
- each neuron stores a weight vector
- neurons compete to represent inputs
- nearby neurons cooperate during learning

Q8- After identifying the winner neuron, the winner and its neighbors are updated toward the input vector.

General update rule:

$$w_i(t + 1) = w_i(t) + \eta(t) h_{ci}(t) [x(t) - w_i(t)]$$

where:

- $w_i(t)$: current weights of neuron i
- $x(t)$: input vector
- $\eta(t)$: learning rate
- $h_{ci}(t)$: neighborhood function relative to winner c

The winner changes the most, neighbors change less, distant neurons may not change.

Q9- A SOM compresses data by replacing many input vectors with a smaller set of representative weight vectors. Each neuron acts like a prototype for similar inputs.

Instead of storing every original input exactly, you can:

- store the SOM codebook vectors
- represent each input by the index of its winning neuron

This reduces storage while preserving the essential structure and similarity relationships of the data. The loss is minimized because each input is represented by the closest prototype.

Q10- A Kohonen network consists of:

- an input layer connected fully to
- a competitive map layer arranged in a geometric grid, usually 2D

Each map neuron has:

- a position on the grid
- a weight vector with the same dimension as the input

There are no target output neurons in the supervised sense. The network organizes itself by competition, neighborhood cooperation, and adaptive weight changes.